A WEAKLY ETALE RING MAP IS FORMALLY UNRAMIFIED

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A pure ideal I satisfies I^2 = I. invoked at proof lines 30426-30427. Makes the quotient I/I^2 vanish.
- The module of differentials of B over A is I/I^2 where I is the kernel of the multiplication map B tensor_A B -> B. invoked at proof lines 30428-30429. Expresses Omega_{B/A} as the vanishing quotient.
- A ring map B over A is formally unramified if and only if Omega_{B/A} = 0. invoked at proof lines 30430-30432. Converts the vanishing into formal unramifiedness.

Source reference. Original definition 04PR: sources/stacks-chow/tags.txt, source lines 6124–6124. Cited at proof lines 30425-30426 to identify the kernel I as a pure ideal; the definition is used inside the counted span.

Source reference. Original lemma 04PS: sources/stacks-chow/tags.txt, source lines 6125–6125. Cited at proof lines 30426-30427 for the idempotence of I; the conclusion is used inside the counted span.

Source reference. Original lemma 00RW: sources/stacks-chow/tags.txt, source lines 1018–1018. Cited at proof lines 30428-30429 to express Omega_{B/A} as I/I^2; the conclusion is used inside the counted span.

Source reference. Original lemma 00UO: sources/stacks-chow/tags.txt, source lines 1118–1118. Cited at proof lines 30430-30432 to convert the vanishing of Omega into formal unramifiedness; the conclusion is used inside the counted span.

Lemma 0.2. *Let $A \to B$ be a ring map such that $B \otimes_A B \to B$ is flat. Then $\Omega_{B/A} = 0$, i.e., B is formally unramified over A.*

Proof. Let $I \subset B \otimes_A B$ be the kernel of the flat surjective map $B \otimes_A B \to B$. Then I is a pure ideal (Algebra, Definition 04PR), so $I^2 = I$ (Algebra, Lemma 04PS). Since $\Omega_{B/A} = I/I^2$ (Algebra, Lemma 00RW) we obtain the vanishing. This means B is formally unramified over A by Algebra, Lemma 00UO. □

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